

The Elliptical Model of Digital Consumption

The geometry of the doomscroll

Lucas Haines

Introduction

TikTok, Instagram, and YouTube are built to hold attention as long as they can. Anyone who has used them knows the rhythm: the first ten minutes feel good, the next hour or two something turns, and by the end you are foggy and irritated and unsure where the time went. The pattern shows up across users and across platforms, which suggests it is not a personal failure but a feature of the design.

The epidemiological evidence points the same direction. In a study of over half a million adolescents, Twenge and colleagues found that those reporting five or more hours of daily device use were 71% more likely than one-hour users to show clinical risk factors for depression and suicide.[1] The question is why scrolling reliably ends in this state, and what shape the experience traces along the way.

This paper argues that a single scrolling session traces an ellipse. The shape divides into four parts that map onto distinct stages of the experience: pleasure, mindless continuation, the crash, and a stretch of numbness afterward. The geometry clarifies why stopping feels so hard, and why the apps are built to push users past the peak rather than let them stop at it.

1. The Map

Picture a graph with two axes. The vertical axis is Q , for Quality, which measures the felt value of the experience. Q above zero is pleasure — the videos land, you feel connected to whoever made them. Q below zero is suffering — anxiety, brain fog, the kind of exhaustion that does not pass with a nap.

The horizontal axis is P , for Phase, which marks position on the ellipse and runs from $-a$ on the left to $+a$ on the right. In the upper half (Q greater than zero), negative P is the approach to the peak, P equal to zero is the peak itself, and positive P is the descent. In the lower half (Q less than zero), the same P values mark the return arc.

A scrolling session moves around this graph in the shape of an ellipse:

$$(P^2 / a^2) + (Q^2 / b^2) = 1$$

Two numbers control the shape. The number a sets the total length of the session. The number b sets the height of the peak.

Why an ellipse, specifically? The model needs a closed curve, one that returns to its starting point, and the ellipse is the simplest such curve specifiable in just two numbers. Other shapes would also work, but they require more parameters without adding analytical value. The ellipse is best read as a stand-in for any smooth, closed shape that rises to a peak and falls back.

Closure is the defining property. Once you have passed into the lower half of the curve within a single session, more scrolling cannot return you to the upper half. The direction of travel along the curve is one-way. You can exit the trajectory at any point by disengaging, but more input cannot reverse the descent.

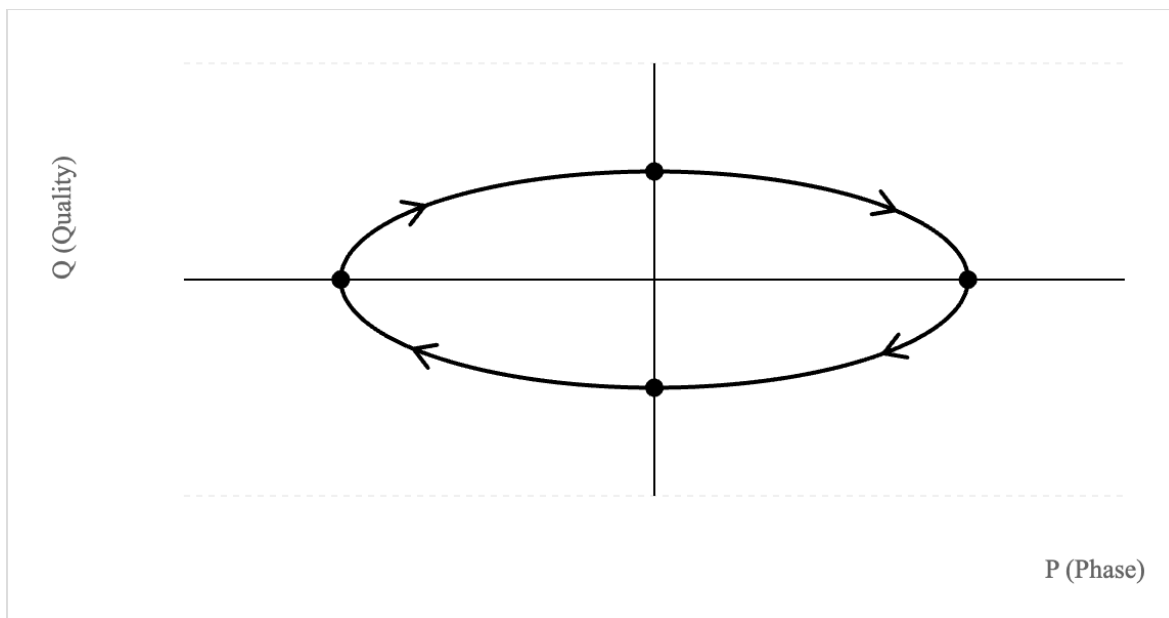


Figure 1. The basic ellipse rises to a peak, falls past zero into the lower half, then loops back to the start.

2. The Four Quadrants

The elliptical trajectory divides into four quadrants, and each one describes a distinct stage of the session.

Quadrant II: The Approach

The upper-left quadrant is the optimal portion of the session. P is negative (still rising), Q is positive (it feels good). This is the first ten or fifteen minutes — the videos feel fresh, the jokes land, stopping seems pointless. You are climbing the steep part of the ellipse.

At the peak, the slope reaches zero. This is where the trick lies. With the slope flat, the next

swipe appears to cost nothing — but the curve has already begun to bend downward, and that next swipe is the first step of the descent.

Wolfram Schultz's work on dopamine neurons offers a mechanism for the misperception.[2] With repeated exposure to a reward, dopamine begins to fire on the anticipation of the reward rather than on its delivery. At the peak of a scrolling session the dopamine release is no longer for the current video but for the expected next one. The brain has already shifted its attention forward; you have not yet felt the shift.

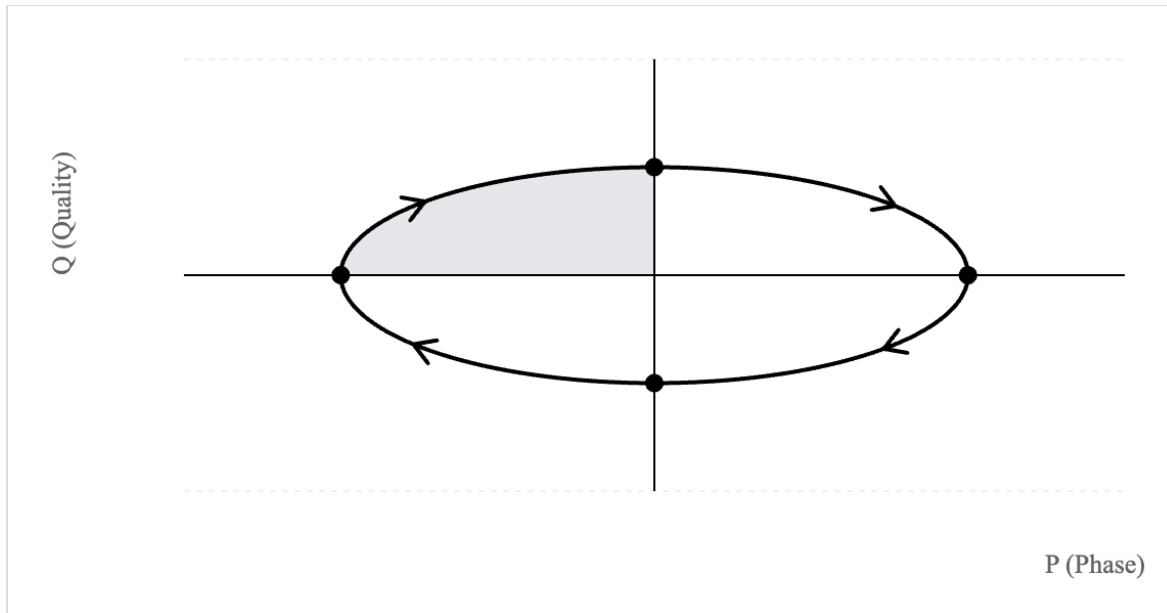


Figure 2. The Approach. Sessions that close out within this upper-left arc remain net positive.

Quadrant I: Diminishing Returns

Past the peak you enter Quadrant I. P is positive, Q is still positive, but the videos no longer engage the way they did. You can sense the change, and you keep scrolling anyway.

Skinner described the initial mechanism: when a reward is unpredictable, animals and people continue to perform the action that occasionally produces it.[3] Slot machines and social media feeds both exploit this directly. As the session deepens, though, the motivation shifts. Natasha Dow Schüll, in her study of gambling addiction, called this later state the “machine zone.”[4] In the machine zone the user's goal is no longer to win a reward (maximizing Q) but to maintain the rhythm of the action itself (continuing along P), as a way to insulate themselves from everything outside the screen. The curve does not drop off a cliff. It bends gently, and the transition from active pleasure to zoned-out compulsion is too smooth for conscious detection.

Quadrant IV: The Crash

Crossing the horizontal axis takes you into Quadrant IV. P is positive, Q is now negative. This is the anxious, foggy state that follows a two-hour TikTok binge — pleasure replaced by a low physical agitation and the need to escape it.

Anna Lembke describes the chemistry of this transition.^[5] Repeated, high-frequency dopamine release prompts the brain to compensate by lowering its baseline; the reward system overshoots while trying to restore balance, and the result is a state of active dopamine debt. The crash is not a moral failure. It is the receipt for the previous hour.

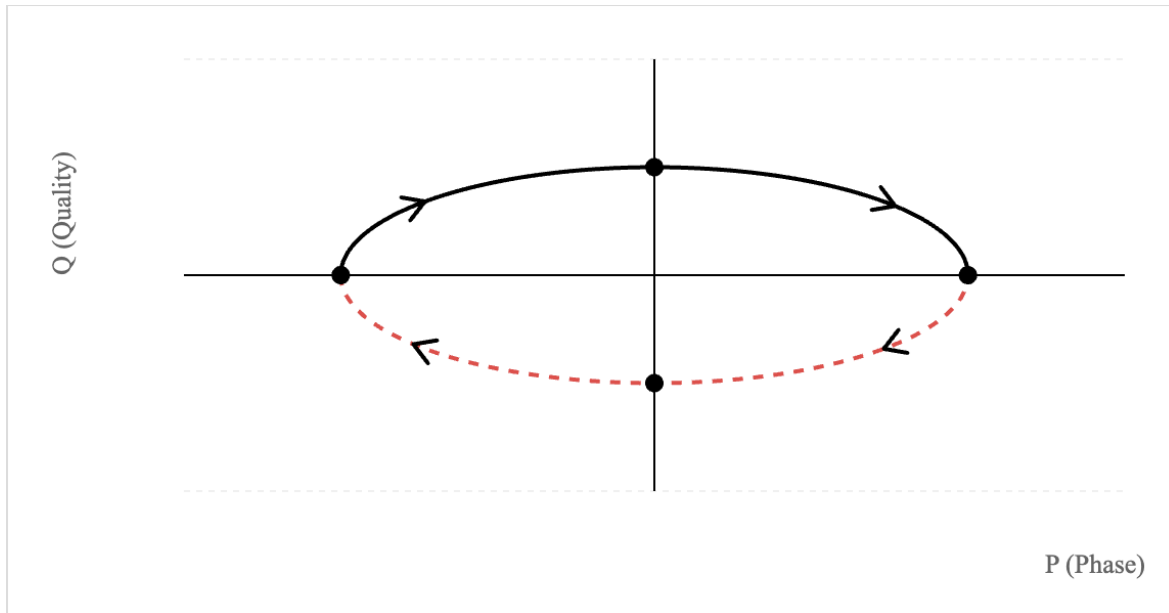


Figure 3. The full trajectory. The dashed arc marks the path most users follow into the lower half.

Quadrant III: Numbness and Closure

The final quadrant is the return arc. You have closed the app, but the session has not ended. Things that normally produce pleasure (a conversation, a sunset, a good book) feel muted. You are moving along the lower-left arc of the ellipse, heading back toward zero.

This is where the closure of the curve does its most important work. While you are in this arc, additional scrolling cannot return the system to pleasure. Effort yields no reward in this region. The way out is to leave, not to continue. Disengagement ends the cycle; continued input only holds you in place.

This is the geometry behind the familiar complaint, “I scrolled for another hour and only felt worse.” The extra hour does not restart the cycle. It holds you in the lower half. The way back to neutral is to step out of the trajectory entirely.

3. Why It Is Hard to Stop

If the peak is identifiable in principle, why do users not stop at it?

In ordinary settings, the slowdown is felt directly. While eating, the stomach signals fullness. In conversation, fatigue gradually settles in and the exchange winds down. The body sends signals that correspond to position on the curve, and the person acts on them.

Apps are designed to interrupt these signals. Infinite scroll and autoplay eliminate the natural stopping point — the next item is always loading. Tristan Harris and the Center for Humane Technology have documented how these “frictionless” features are deliberately engineered to hijack psychological vulnerabilities.[6] Small, low-grade rewards arrive often enough to mask underlying fatigue, and the signal of satiety does not reach you.

There is also a cost to ignoring the signal once it does arrive. James Gross has shown that suppressing an emotion, rather than acting on it, increases physiological stress.[7] When an app encourages you to override the creeping fatigue, you pay for the override as well as for the scrolling.

The work of putting the phone down, then, is not an exercise of heroic willpower. It is the discipline of staying in contact with your actual position on the curve. Once the small dip in interest at the peak becomes noticeable, closing the app follows naturally. The difficulty lies in allowing yourself to feel it.

4. Eccentricity: Short-Form vs. Long-Form Media

The shape of the ellipse depends on the ratio of a to b . Call this ratio E , the eccentricity.

When a is smaller than b , the ellipse is tall and narrow. The peak is high but reached quickly, and the descent is steep. This profile corresponds to short-form video — TikTok, Reels, Shorts. The pleasure is sharp but the optimal zone is brief; the peak arrives within minutes, and the descent begins before the climb has fully registered. Neuroimaging studies of short-form platforms support this, showing that highly personalized, rapid-fire algorithms activate the brain's reward centers aggressively while suppressing prefrontal control networks.[8]] Informatics researcher Gloria Mark has tracked a steady decline in screen attention, with the median duration of focus on a single screen falling from 2.5 minutes in 2004 to 47 seconds in recent measurements.[9] The phase window is narrowing.

When a is larger than b , the ellipse is wide and shallow. The peak is lower but sustained over time, and the decline is gradual. This profile corresponds to long-form media: a podcast, a documentary, a sustained conversation in a group chat. The optimal zone is wide enough that you can remain in it for an hour without entering the descent.

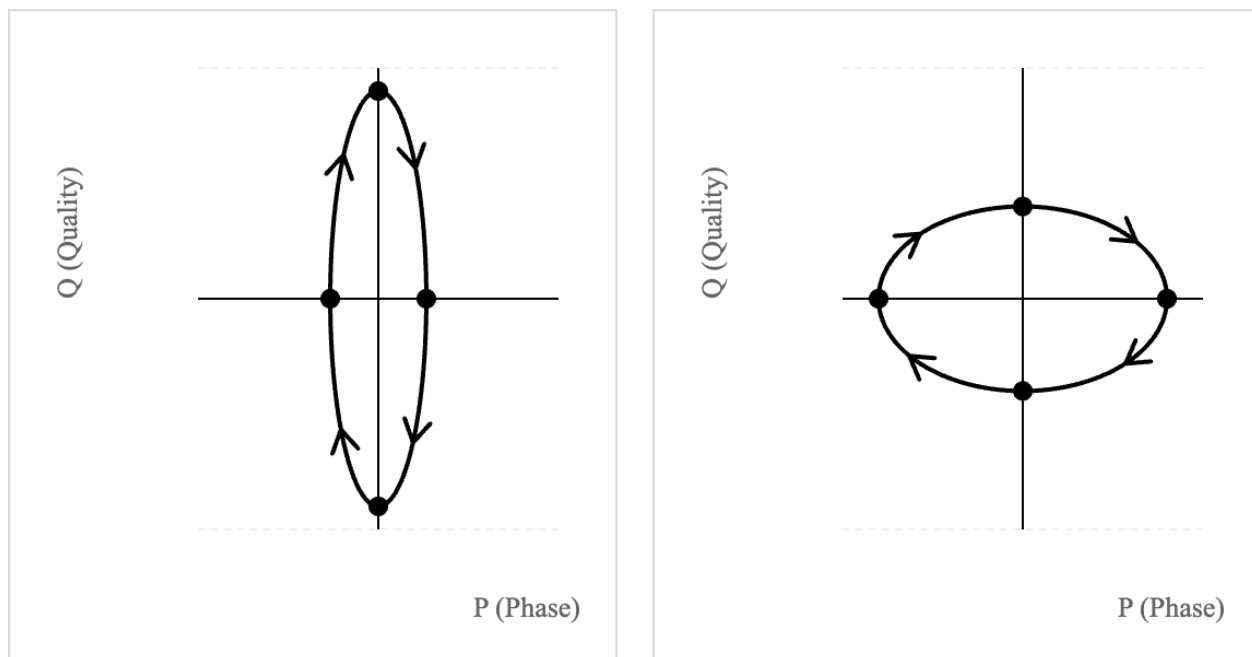


Figure 4a (left). A tall, narrow ellipse — short-form video, brief optimal zone, sharp crash. Figure 4b (right). A wide, shallow ellipse — long-form media, extended optimal zone, gradual decline.

The eccentricity decomposition gives the model empirical content. Two platforms with different E ratios should produce measurably different sessions: a short-form feed produces a sooner, sharper crash than a long-form one.

There is also a middle range. When a and b are close to equal, the ellipse is roughly circular. The peak is moderate, reached at a measured pace, and the descent mirrors the climb. This profile corresponds to media that neither spikes nor sustains — a group text thread, a mid-length YouTube video, a feed with mixed content types. The session ends without a sharp crash, but also without the extended engagement of long-form. For a user optimizing against post-session regret, a near-circular ellipse may be the target geometry.

5. Compounding Across Sessions

Everything above describes a single session in isolation. Each session begins at the leftmost point of the ellipse ($P = -a$, $Q = 0$), traces the curve, and either completes or is exited by disengagement. The next session begins as a new ellipse.

In practice, sessions do not stay independent. With repeated, heavy use, two drifts appear.

The first drift runs along the P axis. The starting point of each new cycle moves forward on the curve. After enough sessions in a short period, the user no longer enters at the bottom of the climb — the first swipes already feel flat. The optimal zone has narrowed, and the user is starting further into the cycle than they were a week ago.

The second drift runs along the Q axis. The entire ellipse sinks: the peak gets lower, the crash gets deeper, and the baseline against which Q is measured has dropped. This long-term downward shift is known in neurobiology as allostatic load. George Koob and Michel Le Moal established that repeated heavy stimulation does not only cause temporary crashes; it persistently lowers the brain's hedonic set-point.^[10] Eventually the starting Q is negative. The user opens the app already in suffering and scrolls to chase a peak that, in absolute terms, never crosses back above zero.

Together the two drifts describe the trajectory of dependence. An ordinary user begins each session at $(-a, 0)$ and traces a normal ellipse. A user who has been heavy for some time begins later on P and lower on Q. The ellipse they trace is shifted down and pushed forward. The fun zone shrinks. In the limit, it disappears entirely.

This is not a separate model — it is the same elliptical structure with a moving origin. Recovery (extended periods of no use) allows both drifts to reverse: the starting P moves back toward $-a$, and the baseline Q rises back toward zero. The geometry stays the same. What changes is where on the page the ellipse is drawn.

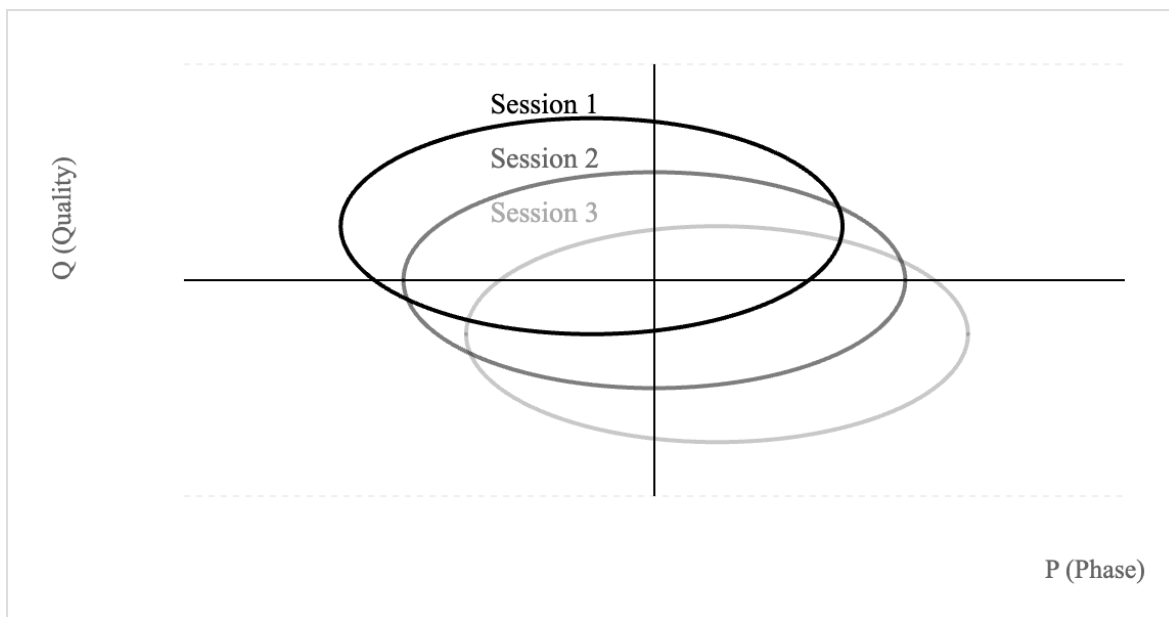


Figure 5. Cumulative drift across sessions. As use accumulates, each new ellipse begins further along P and lower on Q. In severe cases, the entire curve sits below the $Q = 0$ line.

Conclusion

The doomscroll is not a sign of weakness or failed self-control. It is a journey along a closed curve, and the curve has been deliberately engineered to carry you past the peak and through the

lower half.

The elliptical model makes three claims. First, the session has an ideal endpoint — the peak — that is invisible to the person inside it and actively obscured by the platform. Second, once that peak is crossed, no amount of continued input can return the experience to positive territory; the way out of the lower half is to exit, not to persist. Third, across sessions the curve shifts: the fun zone narrows, the baseline drops, and recovery becomes the primary variable.

The model does not argue that digital consumption is inherently harmful. It argues that the current design prevents users from knowing where they are on the curve. A more honest interface would make that position legible — a small friction at the peak, a visible signal that the slope has reversed. The geometry of a better experience is the same; only the information would change.

A more honest design would introduce friction at the peak. A small interruption at the right moment would restore your awareness of position and let the session end naturally. The technology already knows where you are on the curve. The remaining question is whether it is built to let you feel it.

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